

Network Resource Management for Next Generation Applications

Emerging applications such as immersive virtual reality, remotely controlled robots, and IoT sensing promise to revolutionize education, training, healthcare, and agriculture, among other aspects of daily life. However, network communication delay is seen as a key hurdle to enabling these applications [3, 9, 6]. More specifically, such applications are imposing stringent requirements on the network infrastructure in three ways: (i) their ever-increasing demand for more bandwidth (e.g. volumetric video streaming requiring hundreds of Mbps [37]), (ii) their need for low latency (within a few tens to a few hundreds of milliseconds) given their interactive nature [25, 11], and (iii) their increasing scale of deployment, potentially involving hundreds of devices sharing common access points [2, 38].

The last mile network infrastructure, on the other hand, has not been able to keep pace with these requirements. The median broadband speed is still less than 35Mbps in more than half of the countries worldwide [4]! The cellular access bandwidth for a given user typically varies from 5-50Mbps for LTE [64] and 12-200Mbps for 5G [56]. 5G and 6G's promises for high bandwidth with ultra-low latency are yet to be realized. Reducing geographical distance with edge computing can reduce network propagation delays, but cannot reduce the more dominant queuing delays that stem from limited capacity at the last mile.

So the grand challenge we put forward in this section is: *how do we scale and manage the last mile network capacity in order to meet the stringent requirements of emerging networked applications?*

The wireless networking community has been actively working on this challenge over the last few years by innovating on the physical layer to achieve higher bandwidth and scale (e.g. through technologies such as mmWave, massive antenna arrays, beam-forming, etc) [13]. However, we believe that truly meeting this challenge requires a concerted effort from both the wired and the wireless communities. As steps towards tackling this challenge, we outline multiple research directions below that span different aspects of last mile connectivity and require crucial insights and expertise from wired networking.

Endpoint-driven Traffic Shaping on Access Links from the ISP

One seemingly simple way to ensure that interactive time-sensitive applications are not adversely impacted by other contending flows is to prioritize them (e.g. prioritizing a user's Zoom video conferencing or AR/VR traffic over their bulk Dropbox downloads). Even though the basic mechanism for such prioritization was proposed decades ago [59], it is not generally supported by the public Internet today. In particular, such prioritization must be enforced at the bottleneck links where queues build up, which reside within the Internet Service Provider (ISP's) domain. However, ISPs are ill-equipped to unilaterally enforce such prioritization across applications. This is partly because of the philosophical barrier that requires the network to be general and neutral to all applications, partly because of technological barriers stemming from lack of visibility into a user's traffic and their preferences, and partly because of lack of incentives on the ISP's end.

We observe that the philosophical barrier of network neutrality can be removed by allowing the users to control how flows and applications within their *own* traffic bundle are shaped and prioritized with respect to one another. Given that ISPs typically isolate traffic across different users [19], such per-user shaping and prioritization can be very effective, without raising concerns about net neutrality. To that end, mechanisms have been designed that allow users to specify their preferences to the ISPs, such that the ISP can enforce them [34, 18, 21, 46, 65, 8, 7]. However, outside of business agreements established with large enterprises [7, 8], the ISP support for such mechanisms is not available to general customers (e.g. home users and small enterprises), presumably due to lack of incentive and/or scaling limitations.

Some recent mechanisms [20, 63] have therefore been proposed that allow the end-users to unilaterally control how flows within their own traffic bundle are shaped and prioritized *without* any support from the ISP (with Bundler [20] assuming support from the content providers, and CRAB [63] operating solely from the receiver's domain). While these recent efforts show the promise of such an approach, they are limited in several ways. For example, Bundler can only shape flows within traffic bundles from one domain to another (and not across flows sourced from different domains). While CRAB can handle flows from different domains destined to the same receiver, it leads to some inevitable link under-utilization and cannot help short-lived flows. Therefore, more work is needed to understand the fundamental limitations of such approaches and improve their effectiveness. A related question is whether we necessarily need such indirect approaches, or are there ways to enable and incentivize the ISPs to support user-driven prioritization and shaping at scale?

Scaling and Slicing Cellular Infrastructure

Cellular connectivity is becoming increasingly critical for supporting emerging outdoor applications (e.g. IoT devices, drones, and robots in smart-cities and farms). A promising approach for scaling cellular capacity to meet the ever-increasing demands is to augment the coverage regions of cellular base-stations with increasing density of lower powered small cells [57, 17, 13, 52, 15, 51, 1]. Efficient utilization of such dense cellular infrastructure requires tight global coordination between small cells and the base-station (e.g. through coordinated mechanisms for interference management, load balancing, hand-overs, and radio resource scheduling that operate at timescales of hundreds of microseconds to a few milliseconds) [40, 67, 36, 10, 61, 49, 39, 48, 52]. Such tight global coordination requires low latency communication across cells (or between a cell and central controller) [12], along with the ability to perform complex centralized computation quickly at scale. While there have been some initial efforts in these directions (e.g. on scalable signal processing systems [26, 35, 47]), building a practical system of coordinated dense cells remains an interesting open challenge that falls within the purview of wired networking.

Network slicing is another interesting feature introduced in the 5G cellular standards [14, 58, 53, 28, 33], that promises to fulfill the requirements of emerging applications. It refers to dividing cellular network resources among different services or “groups of users” to create virtual customizable networks, referred to as slices. It is envisioned that these slices will enter into service level agreements (SLAs) with the network operator, which will govern the type of service and the total amount of resources allocated to each slice. For example, it can provide ultra-reliable low-latency connectivity to first responders [42] or high throughput connectivity for AR/VR [55]. Moreover, each slice will be able to customize its virtual network and dynamically manage its share of resources, e.g. a digital farming slice can prioritize communication for tele-operated robots over sensor data collection. While the objectives of 5G slicing have been outlined in the standards, the mechanisms of realizing these objectives are still being worked out, with substantial recent work on the topic (e.g. [60, 16, 32, 41, 30, 24, 31, 43, 44, 50, 45, 32, 23, 27, 66]). We believe this is a promising area of research that shares challenges with multi-tenant cloud settings (e.g. dealing with hierarchical fairness across multiple resource types spanning radio access network, edge compute, and the network core; estimating demands of a slice/tenants; mechanisms for specifying service-level agreements and pricing; performance isolation across slices; etc). It can therefore benefit from the insights that the wired networking community has developed over the years in the context of multi-tenant datacenters. Nonetheless, these insights must be adapted to deal with unique challenges that arise in cellular networks, e.g. due to variable wireless channel quality between user device and base-station.

We end this section with pointing out that research on cellular infrastructure has largely been hindered by its closed-sourced nature. However, that landscape is now changing with efforts towards making cellular architecture more open [5, 54] in order to facilitate innovations. Reusing the phrasing from [29], this technology push, combined with the pull stemming from the stringent communication requirements of emerging applications, provides for an opportune timing for more research in this space.

Traffic Management in Emerging Private Wireless Networks

An interesting paradigm that is gaining greater significance in the context of emerging applications is private wireless network, e.g. in factories, farms, smart-cities, etc. A key aspect that differentiates such emerging private network settings from conventional enterprise wireless networks, is greater predictability and control over the network workloads. For example, networked tasks (e.g. a remote tele-operation, sensor data collection, etc) can be actively scheduled based on network availability and application requirements (task deadlines and constraints). We can leverage this greater predictability and control to perform fine-grained centralized traffic engineering in such private wireless network settings. In particular, we can design a cross-layer resource management system that simultaneously optimizes different aspects (task/flow scheduling, routing, and even sending rates) based on global knowledge of application requirements and available network capacity. Such an effort can again extend insights from traffic engineering and flow scheduling in datacenter networks, but must deal with additional challenges stemming from wireless interference and varying signal strength [62].

Application-Driven Networking

More broadly, a key to supporting emerging applications on the last-mile network infrastructure with limited capacity is to incorporate greater *application-awareness* in how the network resources are managed, where we directly and explicitly take specific application requirements into account when deciding how the limited network capacity is used across and within applications. This can range from basic user-driven prioritization of one application over another (§) to application-driven traffic engineering in private networks (§) to more sophisticated mechanisms where the network explicitly favors data packets that carry more meaningful information within a given application (e.g. by dropping

semantically less useful packets during congestion [22]). There are several questions that are interesting to answer in this context: What network-level decisions can and must be informed by application requirements? How do we enable such application-driven network decisions while retaining the network's generality? For instance, can we design parameterized network primitives for making such decisions that the applications and end-users can configure as per their own requirements? How should applications be designed to exploit such network support? Would the capabilities of such application-driven networks be restricted to private settings or can they be extended to public networks?

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